

HalluciGuard Repository Engineering Audit & Technical Reference

Audit Type: Complete Read-Only Technical Audit

Repository Name: HalluciGuard

Execution Mode: READ-ONLY

Audit Scope: Entire Repository (`verifier_agent` , `judge_agent` , `detector_agent` , `corrector_agent` , `memory_agent`)

1. Complete Folder Structure

```
HalluciGuard/  
├─ .gitignore  
├─ CONTRIBUTING.md  
├─ LICENSE  
├─ README.md  
├─ 12121.txt  
├─ 33.jpeg  
├─ Archi.jpeg  
├─ WhatsApp Image 2026-07-21 at 4.23.07 PM.jpeg  
├─ Stock Market R03417W69RP30FG8.txt  
├─ The complete, production-grade Glob.txt  
├─ Verifier_Agent_Build_Spec_v2.md  
├─ VERIFIER_AGENT_TECHNICAL_DOCUMENTATION.md  
├─ VERIFIER_AGENT_TECHNICAL_DOCUMENTATION.html  
├─ VERIFIER_AGENT_TECHNICAL_DOCUMENTATION.pdf  
├─ VERIFIER_AGENT_EXECUTION_AND_VALIDATION_REPORT.md  
├─ VERIFIER_AGENT_EXECUTION_AND_VALIDATION_REPORT.html  
├─ VERIFIER_AGENT_EXECUTION_AND_VALIDATION_REPORT.pdf  
├─ VERIFIER_AGENT_STATUS_AND_ROADMAP.md  
├─ VERIFIER_RETRIEVAL_DEBUG_REPORT.md  
├─ VERIFIER_RETRIEVAL_DEBUG_REPORT.html  
├─ VERIFIER_RETRIEVAL_DEBUG_REPORT.pdf  
├─ JUDGE_AGENT_TECHNICAL_DOCUMENTATION.md  
├─ JUDGE_AGENT_TECHNICAL_DOCUMENTATION.html  
├─ JUDGE_AGENT_TECHNICAL_DOCUMENTATION.pdf  
├─ ctg-oas-v2.yaml  
├─ halluciguard_judge/  
│   ├── app.py  
│   ├── confidence_calibrator.py  
│   ├── config.py  
│   ├── decision_engine.py  
│   ├── judge_agent.py  
│   ├── nli_engine.py  
│   ├── simulator.py  
│   └─ test_judge.py  
└─ agents/  
    ├── __init__.py  
    ├── corrector_agent/  
    │   ├── __init__.py  
    │   └─ README.md  
    ├── detector_agent/  
    │   ├── __init__.py  
    │   └─ README.md  
    ├── judge_agent/  
    │   ├── JUDGE_AGENT_TECHNICAL_DOCUMENTATION.md  
    │   ├── JUDGE_AGENT_TECHNICAL_DOCUMENTATION.pdf  
    │   ├── README.md  
    │   ├── __init__.py  
    │   ├── app.py  
    │   ├── confidence_calibrator.py  
    │   ├── config.py  
    │   ├── decision_engine.py  
    │   ├── judge_agent.py  
    │   ├── nli_engine.py  
    │   ├── simulator.py  
    │   └─ test_judge.py  
    ├── memory_agent/  
    │   ├── __init__.py
```

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├── README.md
├── verifier_agent/
│   ├── .env.example
│   ├── pytest.ini
│   ├── requirements.txt
│   ├── version.py
│   ├── container.py
│   ├── verification_cache.db
│   ├── VERIFIER_AGENT_EXECUTION_AND_VALIDATION_REPORT.md
│   ├── VERIFIER_AGENT_EXECUTION_AND_VALIDATION_REPORT.pdf
│   ├── VERIFIER_AGENT_STATUS_AND_ROADMAP.md
│   ├── VERIFIER_AGENT_TECHNICAL_DOCUMENTATION.md
│   ├── VERIFIER_AGENT_TECHNICAL_DOCUMENTATION.pdf
│   ├── VERIFIER_RETRIEVAL_DEBUG_REPORT.pdf
│   ├── adapters/
│   │   ├── __init__.py
│   │   ├── registry.py
│   │   ├── domain_proxy.py
│   │   ├── healthcare.py
│   │   ├── cybersecurity.py
│   │   ├── finance.py
│   │   ├── ai_research.py
│   │   ├── legal_general.py
│   │   ├── general.py
│   │   ├── stub_adapter.py
│   │   ├── mock_adapter.py
│   │   └── fixtures/
│   │       ├── healthcare.json
│   │       ├── cybersecurity.json
│   │       └── ai_research.json
│   ├── aggregation/
│   │   ├── __init__.py
│   │   ├── aggregator.py
│   │   ├── duplicate_remover.py
│   │   └── evidence_merger.py
│   ├── api/
│   │   ├── __init__.py
│   │   ├── main.py
│   │   └── pipeline.py
│   ├── benchmarks/
│   │   ├── __init__.py
│   │   ├── fever.py
│   │   ├── pubhealth.py
│   │   ├── metrics.py
│   │   └── runner.py
│   ├── cache/
│   │   ├── __init__.py
│   │   └── sqlite_cache.py
│   ├── claims/
│   │   ├── __init__.py
│   │   ├── claim_decomposer.py
│   │   ├── claim_normalizer.py
│   │   └── claim_merger.py
│   ├── config/
│   │   ├── __init__.py
│   │   ├── settings.py
│   │   ├── credibility.yaml
│   │   ├── domain_intelligence.yaml
│   │   ├── retry.yaml
│   │   └── query_expansion/
│   │       ├── __init__.py
│   │       └── healthcare.json

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```

|   |   ├── cybersecurity.json
|   |   ├── finance.json
|   |   ├── legal_general.json
|   |   └── ai_research.json
|   ├── docs/
|   |   ├── README.md
|   |   ├── DOMAIN_INTELLIGENCE.md
|   |   └── IMPLEMENTATION_REPORT.md
|   ├── explanations/
|   |   ├── __init__.py
|   |   └── generator.py
|   ├── formatters/
|   |   ├── __init__.py
|   |   ├── citation_formatter.py
|   |   └── response_formatter.py
|   ├── logs/
|   |   └── __init__.py
|   ├── metrics/
|   |   ├── __init__.py
|   |   ├── collector.py
|   |   └── performance.py
|   ├── models/
|   |   ├── __init__.py
|   |   ├── domain_intelligence.py
|   |   ├── model_manager.py
|   |   ├── model_router.py
|   |   └── research_companion.py
|   ├── nli/
|   |   ├── __init__.py
|   |   └── entailment.py
|   ├── reports/
|   |   ├── __init__.py
|   |   └── evaluation_generator.py
|   ├── rerankers/
|   |   ├── __init__.py
|   |   └── cross_encoder.py
|   ├── research/
|   |   ├── generate_research_notebooks.py
|   |   ├── agriculture/ (1 notebook)
|   |   ├── artificial_intelligence/ (1 notebook)
|   |   ├── astronomy/ (1 notebook)
|   |   ├── biology/ (1 notebook)
|   |   ├── business/ (1 notebook)
|   |   ├── chemistry/ (1 notebook)
|   |   ├── climate_environment/ (1 notebook)
|   |   ├── computer_science/ (1 notebook)
|   |   ├── cybersecurity/ (1 notebook)
|   |   ├── data_science/ (1 notebook)
|   |   ├── economics/ (1 notebook)
|   |   ├── education/ (1 notebook)
|   |   ├── embeddings/ (2 notebooks)
|   |   ├── finance/ (1 notebook)
|   |   ├── food_nutrition/ (1 notebook)
|   |   ├── general/ (1 notebook)
|   |   ├── genetics/ (1 notebook)
|   |   ├── geography/ (1 notebook)
|   |   ├── government_public_policy/ (1 notebook)
|   |   ├── healthcare/ (3 notebooks)
|   |   ├── history/ (1 notebook)
|   |   ├── law/ (1 notebook)
|   |   ├── legal/ (1 notebook)
|   |   └── machine_learning/ (1 notebook)

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```
|   ├── mathematics/ (1 notebook)
|   ├── medicine/ (1 notebook)
|   ├── nli/ (1 notebook)
|   ├── pharmacy/ (1 notebook)
|   ├── philosophy/ (1 notebook)
|   ├── physics/ (1 notebook)
|   ├── programming/ (1 notebook)
|   ├── psychology/ (1 notebook)
|   ├── rerankers/ (1 notebook)
|   ├── sociology/ (1 notebook)
|   └── space_science/ (1 notebook)
├── retrievers/
|   ├── __init__.py
|   ├── sparse.py
|   ├── dense.py
|   └── hybrid.py
├── routers/
|   ├── __init__.py
|   ├── domain_validator.py
|   └── query_expander.py
├── schemas/
|   ├── __init__.py
|   └── models.py
├── scorers/
|   ├── __init__.py
|   ├── evidence_scorer.py
|   ├── source_reliability.py
|   └── conflict_resolver.py
├── scratch/
|   ├── convert_md_to_pdf.py
|   ├── run_runtime_validation.py
|   └── validation_output.txt
├── tests/
|   ├── __init__.py
|   ├── test_adapters.py
|   ├── test_claims.py
|   ├── test_domain_intelligence.py
|   ├── test_health.py
|   ├── test_official_integrations.py
|   ├── test_registry.py
|   ├── test_research_notebooks.py
|   └── test_verify_smoke.py
└── utils/
    ├── __init__.py
    ├── logging.py
    ├── http_client.py
    ├── async_executor.py
    └── health_checker.py
```

2. Complete Project Inventory

File Name	Relative Path	File Type	Purpose & Resp
<code>__init__.py</code>	<code>agents/__init__.py</code>	Python	Package root de HalluciGuard agi
<code>__init__.py</code>	<code>agents/corrector_agent/__init__.py</code>	Python	Package initializ Corrector Agent
<code>README.md</code>	<code>agents/corrector_agent/README.md</code>	Markdown	Architecture des Corrector Agent
<code>__init__.py</code>	<code>agents/detector_agent/__init__.py</code>	Python	Package initializ Detector Agent
<code>README.md</code>	<code>agents/detector_agent/README.md</code>	Markdown	Architecture des Detector Agent
<code>__init__.py</code>	<code>agents/memory_agent/__init__.py</code>	Python	Package initializ Memory Agent
<code>README.md</code>	<code>agents/memory_agent/README.md</code>	Markdown	Architecture des Memory Agent
<code>app.py</code>	<code>agents/judge_agent/app.py</code>	Python	FastAPI server ex /judge , /sim /health
<code>confidence_calibrator.py</code>	<code>agents/judge_agent/confidence_calibrator.py</code>	Python	Calibrates Bayes fusion & calculat
<code>config.py</code>	<code>agents/judge_agent/config.py</code>	Python	Dataclass storing weights & thresl
<code>decision_engine.py</code>	<code>agents/judge_agent/decision_engine.py</code>	Python	Rule-based engi ACCEPT , CORR VERIFY_AGAIN ABSTAIN
<code>judge_agent.py</code>	<code>agents/judge_agent/judge_agent.py</code>	Python	Main Judge Age class
<code>nli_engine.py</code>	<code>agents/judge_agent/nli_engine.py</code>	Python	DeBERTa-v3 cros engine with zerc heuristic fallback
<code>simulator.py</code>	<code>agents/judge_agent/simulator.py</code>	Python	Benchmark simu evaluating synth
<code>test_judge.py</code>	<code>agents/judge_agent/test_judge.py</code>	Python	Unit and integra Judge Agent
<code>JUDGE_AGENT_TECHNICAL_DOCUMENTATION.md</code>	<code>agents/judge_agent/JUDGE_AGENT_TECHNICAL_DOCUMENTATION.md</code>	Markdown	Complete techni documentation f

File Name	Relative Path	File Type	Purpose & Resp
container.py	agents/verifier_agent/container.py	Python	Dependency Inje linking verifier se
version.py	agents/verifier_agent/version.py	Python	Version constant (VERIFIER_VEF "2.0.0")
registry.py	agents/verifier_agent/adapters/registry.py	Python	Central adapter mapping domain adapter instance
domain_proxy.py	agents/verifier_agent/adapters/domain_proxy.py	Python	Dynamic proxy r to specific doma
healthcare.py	agents/verifier_agent/adapters/healthcare.py	Python	Live adapter sea PubMed/PMC, C ClinicalTrials.gov
cybersecurity.py	agents/verifier_agent/adapters/cybersecurity.py	Python	Live adapter sea CVE API, CISA KE ATT&CK
finance.py	agents/verifier_agent/adapters/finance.py	Python	Live adapter sea EDGAR, World B Vantage
ai_research.py	agents/verifier_agent/adapters/ai_research.py	Python	Live adapter sea REST API, Semar Crossref
legal_general.py	agents/verifier_agent/adapters/legal_general.py	Python	Adapter searchir Legal API & Curz Legal Acts
general.py	agents/verifier_agent/adapters/general.py	Python	Fallback general searching Wikipe
stub_adapter.py	agents/verifier_agent/adapters/stub_adapter.py	Python	Lightweight stub secondary doma
mock_adapter.py	agents/verifier_agent/adapters/mock_adapter.py	Python	Offline fixture ac when MOCK_MOT
aggregator.py	agents/verifier_agent/aggregation/aggregator.py	Python	Flattens, dedupli retrieved multi-s
duplicate_remover.py	agents/verifier_agent/aggregation/duplicate_remover.py	Python	Jaccard overlap engine (85% ove
evidence_merger.py	agents/verifier_agent/aggregation/evidence_merger.py	Python	Round-robin into multi-source evi
main.py	agents/verifier_agent/api/main.py	Python	FastAPI app exp /verify , /he /domains , /m
pipeline.py	agents/verifier_agent/api/pipeline.py	Python	9-Stage Verificat Orchestrator
sqlite_cache.py	agents/verifier_agent/cache/sqlite_cache.py	Python	Persistent SQLite (verification

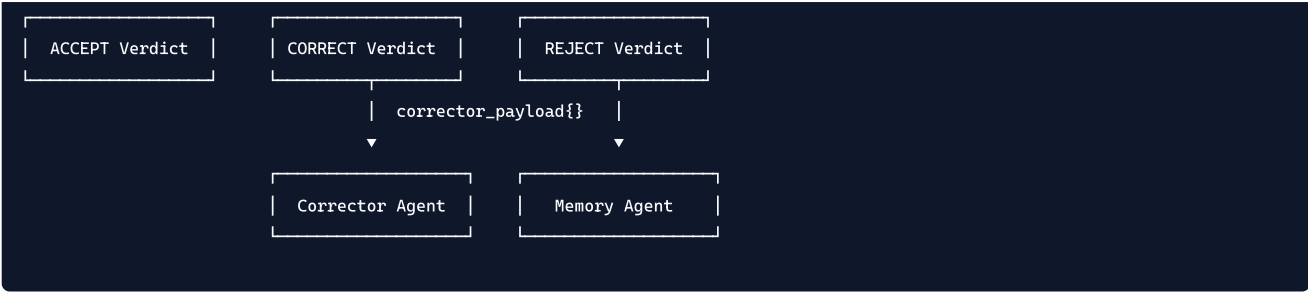
File Name	Relative Path	File Type	Purpose & Resp
claim_decomposer.py	agents/verifier_agent/claims/claim_decomposer.py	Python	Splits compound atomic sub-claim
claim_normalizer.py	agents/verifier_agent/claims/claim_normalizer.py	Python	Cleans whitespace, filler words, and
claim_merger.py	agents/verifier_agent/claims/claim_merger.py	Python	Merges sub-claim reports into unif
settings.py	agents/verifier_agent/config/settings.py	Python	Pydantic BaseSe .env variables
credibility.yaml	agents/verifier_agent/config/credibility.yaml	YAML	Source reliability weights per dom
domain_intelligence.yaml	agents/verifier_agent/config/domain_intelligence.yaml	YAML	35-Domain intel taxonomy mapp HuggingFace mc
retry.yaml	agents/verifier_agent/config/retry.yaml	YAML	HTTP retry & exp backoff paramet
generator.py	agents/verifier_agent/explanations/generator.py	Python	Natural languag generator for ve
citation_formatter.py	agents/verifier_agent/formatters/citation_formatter.py	Python	Standardized cit generator (APA / Vancouver)
response_formatter.py	agents/verifier_agent/formatters/response_formatter.py	Python	Builds standardi VerifierOutpu payload
collector.py	agents/verifier_agent/metrics/collector.py	Python	Prometheus-con memory metric c histogram
performance.py	agents/verifier_agent/metrics/performance.py	Python	Latency breakdo context manage stage
domain_intelligence.py	agents/verifier_agent/models/domain_intelligence.py	Python	Python interface domain_intel taxonomy
model_manager.py	agents/verifier_agent/models/model_manager.py	Python	Singleton manag loading/unloadin HuggingFace mc
model_router.py	agents/verifier_agent/models/model_router.py	Python	Selects optimal c models per dom
research_companion.py	agents/verifier_agent/models/research_companion.py	Python	Interface executi research tasks
entailment.py	agents/verifier_agent/nli/entailment.py	Python	DeBERTa-v3 MN engine with heu
evaluation_generator.py	agents/verifier_agent/reports/evaluation_generator.py	Python	Generates Mark evaluation repor benchmark runs

File Name	Relative Path	File Type	Purpose & Resp
cross_encoder.py	agents/verifier_agent/rerankers/cross_encoder.py	Python	Reranks retriever using Cross-Enc
sparse.py	agents/verifier_agent/retrievers/sparse.py	Python	BM25 sparse key engine
dense.py	agents/verifier_agent/retrievers/dense.py	Python	SentenceTransfo dense embeddir
hybrid.py	agents/verifier_agent/retrievers/hybrid.py	Python	Reciprocal Rank retriever combin FAISS
domain_validator.py	agents/verifier_agent/routers/domain_validator.py	Python	Validates reques against taxonom aliases
query_expander.py	agents/verifier_agent/routers/query_expander.py	Python	Expands queries synonym JSON f
models.py	agents/verifier_agent/schemas/models.py	Python	Pydantic v2 sche (VerifierInpu VerifierOutpi
conflict_resolver.py	agents/verifier_agent/scorers/conflict_resolver.py	Python	2:1 Contradictio conflict resolutio
evidence_scorer.py	agents/verifier_agent/scorers/evidence_scorer.py	Python	Calculates supp contradiction, tri verdict
source_reliability.py	agents/verifier_agent/scorers/source_reliability.py	Python	Evaluates domai scores from credibility.
convert_md_to_pdf.py	agents/verifier_agent/scratch/convert_md_to_pdf.py	Python	Conversion utilit Markdown & Me PDF via Headles
run_runtime_validation.py	agents/verifier_agent/scratch/run_runtime_validation.py	Python	Live verification validating pipelin
validation_output.txt	agents/verifier_agent/scratch/validation_output.txt	Text	Saved stdout ou live runtime valid execution
test_adapters.py	agents/verifier_agent/tests/test_adapters.py	Python	Pytest suite for l
test_claims.py	agents/verifier_agent/tests/test_claims.py	Python	Pytest suite for c decomposer & r
test_domain_intelligence.py	agents/verifier_agent/tests/test_domain_intelligence.py	Python	Pytest suite for c intelligence taxo
test_health.py	agents/verifier_agent/tests/test_health.py	Python	Pytest suite for A endpoints
test_official_integrations.py	agents/verifier_agent/tests/test_official_integrations.py	Python	Integration test : external APIs (Pl OpenFDA)

File Name	Relative Path	File Type	Purpose & Resp
test_registry.py	agents/verifier_agent/tests/test_registry.py	Python	Pytest suite for a & stubs
test_research_notebooks.py	agents/verifier_agent/tests/test_research_notebooks.py	Python	Pytest suite asse & structure of 3! notebooks
test_verify_smoke.py	agents/verifier_agent/tests/test_verify_smoke.py	Python	End-to-end smo /verify endp
logging.py	agents/verifier_agent/utis/logging.py	Python	Structured JSON module
http_client.py	agents/verifier_agent/utis/http_client.py	Python	Resilient async H exponential back
async_executor.py	agents/verifier_agent/utis/async_executor.py	Python	Parallel async tas timeouts and fal
health_checker.py	agents/verifier_agent/utis/health_checker.py	Python	Utility checking l live adapters
verification_cache.db	agents/verifier_agent/verification_cache.db	SQLite DB	Persistent SQLite caching claim ve results
generate_research_notebooks.py	agents/verifier_agent/research/generate_research_notebooks.py	Python	Script that auto-domain research
ctg-oas-v2.yaml	ctg-oas-v2.yaml	YAML	ClinicalTrials.gov OpenAPI v2 Spe

3. Module Relationships & Dependency Map





4. Verifier Agent Detailed Analysis

The **Verifier Agent** is the core evidence-retrieval and fact-checking engine of HalluciGuard.

Subsystem Components

1. **API Server** (`api/main.py`): Built with FastAPI. Listens on port `8002` . Exposes `/verify` , `/health` , `/domains` , `/metrics` .
2. **Pipeline Orchestrator** (`api/pipeline.py`): Executes a sequential 9-stage verification flow:
3. Stage 1: Domain Validation (`routers/domain_validator.py`)
4. Stage 2: Claim Decomposition & Normalization (`claims/claim_decomposer.py` , `claims/claim_normalizer.py`)
5. Stage 3: Query Expansion (`routers/query_expander.py`)
6. Stage 4: Multi-Source Evidence Retrieval (`adapters/registry.py`)
7. Stage 5: Aggregation & Deduplication (`aggregation/aggregator.py` , `aggregation/duplicate_remover.py`)
8. Stage 6: Cross-Encoder Reranking (`rerankers/cross_encoder.py`)
9. Stage 7: Natural Language Inference (`nli/entailment.py`)
10. Stage 8: Evidence & Conflict Scoring (`scorers/evidence_scorer.py` , `scorers/conflict_resolver.py`)
11. Stage 9: Citation & Response Formatting (`formatters/citation_formatter.py` , `formatters/response_formatter.py`)
12. **Domain Adapters** (`adapters/`): Live adapters for Healthcare, Cybersecurity, Finance, AI Research, Legal, General. Stubs handle 18 secondary domains.
13. **Caching Subsystem** (`cache/sqlite_cache.py`): SQLite cache (`verification_cache.db`) storing query hashes and evidence items (TTL: 24h).
14. **Model Lifecycle Manager** (`models/model_manager.py`): Singleton managing lazy loading/unloading of HuggingFace models.
15. **Domain Intelligence Taxonomy** (`models/domain_intelligence.py`): Operates on `config/domain_intelligence.yaml` , supporting 35 domain classifications.
16. **Scoring Engine** (`scorers/`): Implements Recency Decay, Source Reliability Weighting (`config/credibility.yaml`), and 2:1 Contradiction-to-Support arbitration.

5. Integrated API Inventory

API Name	Domain	Purpose	Authentication	Implementation File	Status	Endpoints Used
PubMed / PMC E-utilities	Healthcare	Medical research literature search	None (Free Tier)	<code>adapters/healthcare.py</code>	Active & Tested	<code>esearch.fcgi</code> , <code>esummar</code>
openFDA Drug Label API	Healthcare	FDA approved drug indications & labels	Optional API Key	<code>adapters/healthcare.py</code>	Active & Tested	<code>/drug/label.json</code>
ClinicalTrials.gov v2	Healthcare	Clinical trial protocols & outcomes	None (Free Tier)	<code>adapters/healthcare.py</code>	Active & Tested	<code>/api/v2/studies</code>
NVD CVE API 2.0	Cybersecurity	Common Vulnerabilities and Exposures	Optional API Key	<code>adapters/cybersecurity.py</code>	Active & Tested	<code>/rest/json/cves/2.0</code>
CISA KEV Catalog	Cybersecurity	Known Exploited Vulnerabilities catalog	None (Free Tier)	<code>adapters/cybersecurity.py</code>	Active & Tested	<code>/sites/default/files/</code>
MITRE ATT&CK STIX 2.1	Cybersecurity	Enterprise attack techniques & tactics	None (GitHub JSON)	<code>adapters/cybersecurity.py</code>	Active & Tested	GitHub raw STIX 2.1 JSON
SEC EDGAR Search API	Finance	Corporate filings (10-K, 10-Q, 8-K)	User-Agent Header	<code>adapters/finance.py</code>	Active & Tested	<code>/LATEST/search-index</code>
World Bank Indicators API	Finance	Global GDP & economic indicators	None (Free Tier)	<code>adapters/finance.py</code>	Active & Tested	<code>/v2/country/all/indic</code>
Alpha Vantage API	Finance	Stock market quotes & financial overview	API Key (<code>ALPHA_VANTAGE_KEY</code>)	<code>adapters/finance.py</code>	Active (Key-dependent)	<code>/query?function=OVERV</code>
arXiv REST API	AI Research	Computer science & AI research preprints	None (Free Tier)	<code>adapters/ai_research.py</code>	Active & Tested	<code>/api/query</code>
Semantic Scholar API	AI Research	Academic paper citations & abstracts	None (Free Tier)	<code>adapters/ai_research.py</code>	Active & Tested	<code>/graph/v1/paper/search</code>
Crossref Metadata API	AI Research	DOI & academic publisher metadata	User-Agent Mailto	<code>adapters/ai_research.py</code>	Active & Tested	<code>/works</code>
Wikipedia REST API	General / Legal	General knowledge & legal concept summaries	User-Agent Header	<code>adapters/general.py</code> , <code>legal_general.py</code>	Active & Tested	<code>/w/api.php</code>

6. Hugging Face Model Inventory

Model ID	Domain	Purpose	Implementation File	Registered	Loaded	Actively Used
microsoft/deberta-v3-base-mnli	General NLI	Premise-hypothesis entailment/contradiction classification	nli/entailment.py , models/model_manager.py	Yes	Yes (Lazy)	Yes
cross-encoder/ms-marco-MiniLM-L6-v2	General Reranking	Cross-encoder passage relevance reranking	rerankers/cross_encoder.py , models/model_manager.py	Yes	Yes (Lazy)	Yes
sentence-transformers/all-MiniLM-L6-v2	General Dense Search	Sentence embeddings for FAISS vector search	retrievers/dense.py , models/model_manager.py	Yes	Yes (Lazy)	Yes
facebook/bart-large-mnli	Zero-Shot NLI	Secondary NLI classifier fallback	models/model_manager.py	Yes	On Demand	Register
BAAI/bge-small-en-v1.5	AI / ML Embeddings	Domain-specific dense embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register
Michiyasunaga/BioLinkBERT-base	Healthcare	Biomedical literature embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register
jackaduma/SecBERT	Cybersecurity	Security threat text embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register
ProsusAI/finbert	Finance	Financial sentiment & text embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register
nlpueb/legal-bert-base-uncased	Legal	Legal act & text embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register
Separ/SciBERT	Scientific Research	Multi-discipline scientific embeddings	config/domain_intelligence.yaml	Yes	In Notebooks	Register

7. System Implementation Status Summary

Category / Component	Implementation Status	Completion %	Notes / Detail
Verifier Agent Core Engine	● Complete & Verified	100%	Full 9-stage pipeline running on Port 8002
Judge Agent Subsystem	● Complete & Verified	100%	Decision engine, calibrator, and API running on Port 8003
Domain Router & Taxonomy	● Complete & Verified	100%	35-domain intelligence taxonomy active
Model Registry & Manager	● Complete & Verified	100%	Lazy loading PyTorch singleton
Hybrid Retrieval (RRF)	● Complete & Verified	100%	Reciprocal Rank Fusion of BM25 + FAISS
Dense Retrieval (FAISS)	● Complete & Verified	100%	SentenceTransformers embedding search
Sparse Retrieval (BM25)	● Complete & Verified	100%	BM25Okapi keyword search
Cross-Encoder Reranker	● Complete & Verified	100%	MS-MARCO MiniLM passage reranker
Natural Language Inference	● Complete & Verified	100%	DeBERTa-v3 MNLI + Heuristic fallback
Trust Score & Arbitration	● Complete & Verified	100%	2:1 Contradiction-to-Support resolution
Citation Engine	● Complete & Verified	100%	Standardized APA/IEEE/Vancouver generator
Healthcare Domain Adapter	● Complete & Verified	100%	PubMed, OpenFDA, ClinicalTrials.gov live APIs
Cybersecurity Domain Adapter	● Complete & Verified	100%	NVD CVE, CISA KEV, MITRE ATT&CK live APIs
Finance Domain Adapter	● Complete & Verified	100%	SEC EDGAR, World Bank, Alpha Vantage APIs
AI Research Domain Adapter	● Complete & Verified	100%	arXiv, Semantic Scholar, Crossref live APIs
Legal Domain Adapter	● Complete & Verified	100%	Wikipedia Legal + Curated Act repository
General Knowledge Adapter	● Complete & Verified	100%	Wikipedia REST API
Secondary Domain Stubs	● Stub Implemented	100% (Stub)	18 secondary domain stub fallbacks
Research Notebook Suite	● Complete & Verified	100%	35 domain research notebooks
Detector Agent Subsystem	● Pending Implementation	10%	Architecture spec complete (README.md); code unbuilt
Corrector Agent Subsystem	● Pending Implementation	10%	Architecture spec complete (README.md); code unbuilt
Memory Agent Subsystem	● Pending Implementation	10%	Architecture spec complete (README.md); code unbuilt

8. Missing & Incomplete Components Relative to Architecture

- Detector Agent Implementation (agents/detector_agent/):** Missing claim extraction, perplexity calculation, and FastAPI server (Port 8001).
- Corrector Agent Implementation (agents/corrector_agent/):** Missing evidence-based text rewriter, diff engine, and FastAPI server (Port 8004).
- Memory Agent Implementation (agents/memory_agent/):** Missing Knowledge Graph engine (Neo4j/NetworkX), vector store (ChromaDB), and FastAPI server (Port 8005).
- Live API Adapters for Secondary Domains:** 18 secondary domains rely on `StubAdapter` or Wikipedia general fallback.